

Software Requirements Specification (SRS)

CampusConnect AI

Intelligent Communication, Collaboration & Discussion Portal for College Management

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1. Introduction

1.1 Project Overview

Educational institutions often rely on multiple communication platforms such as WhatsApp groups, emails, physical notice boards, and classroom announcements. These fragmented channels make it difficult for students and faculty to access accurate information, resulting in missed announcements, duplicate queries, poor collaboration, and inefficient communication.

CampusConnect AI is a centralized web-based platform designed to streamline communication and collaboration within a college. The system integrates department-specific communities, academic discussion forums, AI-assisted notice search, smart study groups, lost & found services, anonymous feedback, emergency alerts, and user feedback into a single secure portal.

The platform enhances transparency, improves academic collaboration, and enables efficient information sharing among students, faculty, and administrators.

2. Problem Statement

Current communication methods in colleges suffer from several challenges:

- Information is scattered across multiple platforms.
- Important announcements are often missed.
- Students repeatedly ask the same questions.
- There is no centralized academic discussion platform.
- Study group formation is difficult.
- Lost items are managed manually.
- Feedback collection lacks organization and transparency.
- Emergency communication is slow and inconsistent.

These issues reduce productivity and affect the overall student experience.

3. Proposed Solution

CampusConnect AI addresses these challenges by providing a centralized communication and collaboration platform for educational institutions.

The system enables students, faculty, and administrators to interact through department-specific communities and subject discussion forums while offering AI-assisted search, real-time notifications, verified responses, study group management, lost & found reporting, anonymous feedback, and emergency broadcasting.

By consolidating all communication into one platform, CampusConnect AI improves accessibility, collaboration, and operational efficiency.

4. Objectives

The primary objectives of the system are:

- Centralize all academic communication.
 - Improve collaboration between students and faculty.
 - Reduce duplicate discussions using AI.
 - Enable secure role-based communication.
 - Improve access to academic resources.
 - Digitize lost and found management.
 - Provide instant emergency communication.
 - Collect feedback to continuously improve the platform.
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5. Scope

The proposed system is intended for educational institutions and will be used by:

- Students
- Faculty Members
- Department Coordinators
- Administrators
- Student Clubs
- Placement Cell

The system supports communication across academic and administrative activities while maintaining secure access through role-based authentication.

6. User Roles

Administrator

The administrator is responsible for managing the entire platform.

Responsibilities include:

- Managing users
 - Managing departments
 - Managing discussion categories
 - Sending emergency alerts
 - Viewing analytics
 - Reviewing anonymous feedback
 - Moderating discussions
 - Managing notices
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Faculty

Faculty members can:

- Create announcements
 - Participate in discussions
 - Verify correct answers
 - Moderate subject discussions
 - Guide study groups
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Student

Students can:

- Create discussions
 - Ask academic questions
 - Join study groups
 - Report lost or found items
 - Submit anonymous feedback
 - Rate the platform
 - Receive notifications
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7. Functional Requirements

Module 1 – User Authentication

The system shall:

- Register users
 - Authenticate users securely
 - Support login and logout
 - Manage user profiles
 - Implement role-based access
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Module 2 – AI Smart Notice Board

The system shall:

- Display categorized announcements
 - Support keyword and AI-based searches
 - Filter notices by department and category
 - Show recent and important announcements
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Module 3 – Department Communities

The system shall:

- Create department-specific communities
 - Allow department discussions
 - Restrict access based on department
 - Share department announcements
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Module 4 – Subject Discussion Rooms

The system shall:

- Create subject-wise discussion forums
 - Allow questions and replies
 - Upload supporting files
 - Search discussions
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Module 5 – Verified Answers

The system shall:

- Allow faculty to verify answers
 - Highlight accepted solutions
 - Display official responses separately
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Module 6 – AI Duplicate Question Detection

Before publishing a discussion, the system shall:

- Compare the new question with existing discussions
 - Suggest similar questions
 - Encourage users to continue existing discussions
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Module 7 – Study Group Finder

The system shall:

- Allow users to create study groups
 - Join existing groups
 - Search groups by subject
 - Manage group members
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Module 8 – Lost & Found Portal

The system shall:

- Report lost items
 - Report found items
 - Upload item images
 - Track claim status
 - Search reported items
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Module 9 – Anonymous Feedback

The system shall:

- Allow students to submit anonymous complaints and suggestions
 - Protect the identity of the sender
 - Allow only administrators to access submitted feedback
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Module 10 – Emergency Alert System

The administrator shall be able to:

- Broadcast emergency announcements
 - Notify all users instantly
 - Display alerts prominently on the dashboard
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Module 11 – Discussion Workflow

Every discussion shall have a status:

- Open
- In Progress
- Answered
- Resolved
- Closed

Faculty and administrators may update the status to improve discussion tracking.

Module 12 – User Feedback & Rating

Users shall be able to:

- Rate the platform
- Submit suggestions
- Report bugs
- Recommend new features

Administrators shall be able to review and analyze all submitted feedback.

Module 13 – Real-Time Notifications

Users shall receive notifications for:

- New announcements
 - Replies to discussions
 - Study group invitations
 - Emergency alerts
 - Verified answers
 - Lost item updates
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8. Non-Functional Requirements

Performance

- Response time should be less than 3 seconds under normal usage.
- The platform should support hundreds of concurrent users.

Security

- Passwords must be securely encrypted.
- JWT-based authentication shall be implemented.
- Role-based authorization shall protect sensitive modules.
- Input validation shall prevent SQL injection and cross-site scripting attacks.

Reliability

- The system should maintain high availability.
- Proper error handling and logging should be implemented.

Scalability

The architecture should support future expansion to accommodate additional departments, users, and new modules.

Usability

- Responsive design
 - Simple navigation
 - User-friendly interface
 - Mobile-compatible layout
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9. Technology Stack

Component	Technology
Frontend	React.js
Styling	Tailwind CSS
Backend	Node.js
Framework	Express.js
Database	MySQL
Authentication	JWT & bcrypt
Real-Time Communication	Socket.IO
AI Services	OpenAI API / Gemini API
Version Control	Git & GitHub

10. System Architecture

The architecture follows a three-tier model.

Presentation Layer

- React.js
- Tailwind CSS



Application Layer

- Node.js
- Express.js
- JWT Authentication
- Socket.IO
- AI Integration



Data Layer

- MySQL Database

This architecture ensures modularity, scalability, and secure communication between the client and server.

11. Key Features

- AI Smart Notice Board
 - Department-Based Communities
 - Subject Discussion Rooms
 - Verified Faculty Answers
 - AI Duplicate Question Detection
 - Study Group Finder
 - Lost & Found Portal
 - Anonymous Feedback
 - Emergency Alert System
 - Discussion Workflow Management
 - User Feedback & Rating
 - Real-Time Notifications
 - Role-Based Dashboard
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12. Expected Outcomes

The proposed system is expected to:

- Improve communication efficiency.
 - Increase academic collaboration.
 - Reduce duplicate discussions.
 - Deliver important announcements effectively.
 - Provide secure communication.
 - Enhance student engagement.
 - Improve administrative efficiency.
 - Create a centralized knowledge-sharing platform.
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13. Future Enhancements

The system can be further enhanced with:

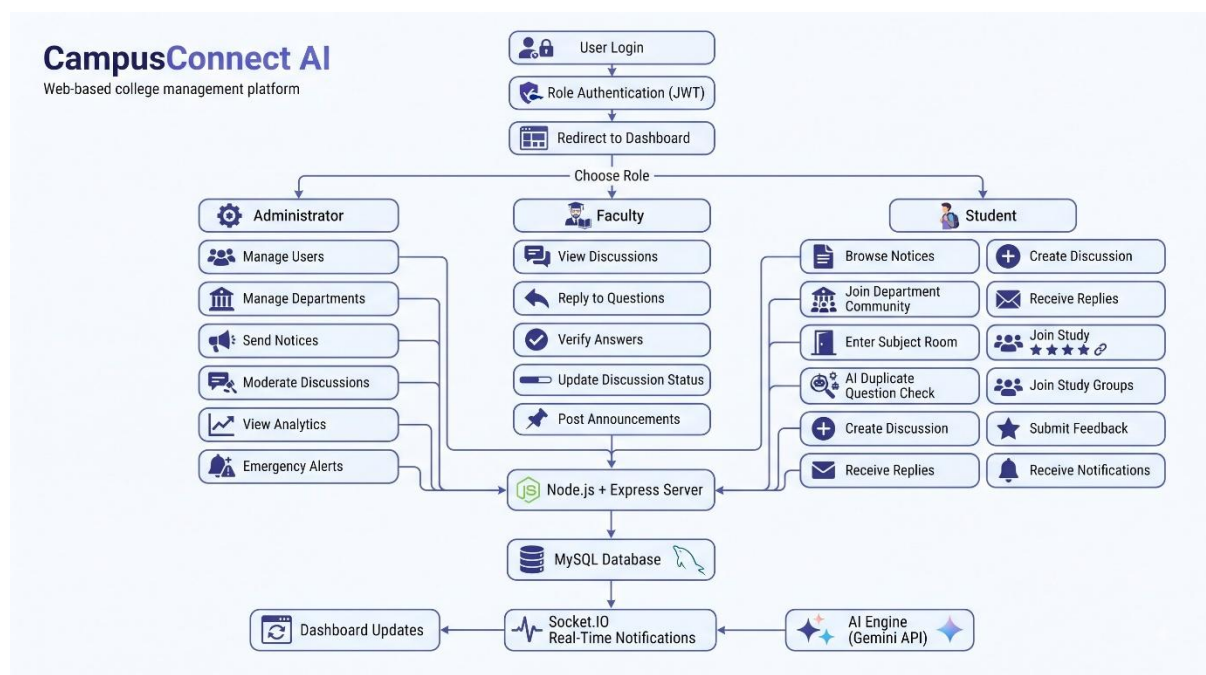
- Mobile application
- AI-generated discussion summaries
- QR-based event attendance

- Calendar integration
- Placement management
- LMS integration
- Multi-language support
- Push notifications

14. Conclusion

CampusConnect AI is an intelligent communication and collaboration platform designed specifically for educational institutions. By combining AI-assisted features, structured discussion forums, role-based access, and real-time communication into a single application, the system addresses the limitations of traditional communication methods.

The proposed solution improves transparency, encourages knowledge sharing, enhances collaboration, and provides an efficient digital ecosystem for students, faculty, and administrators. Its scalable architecture and modern technology stack make it suitable for deployment in colleges seeking a secure and intelligent communication platform.



Student Workflow - CampusConnect



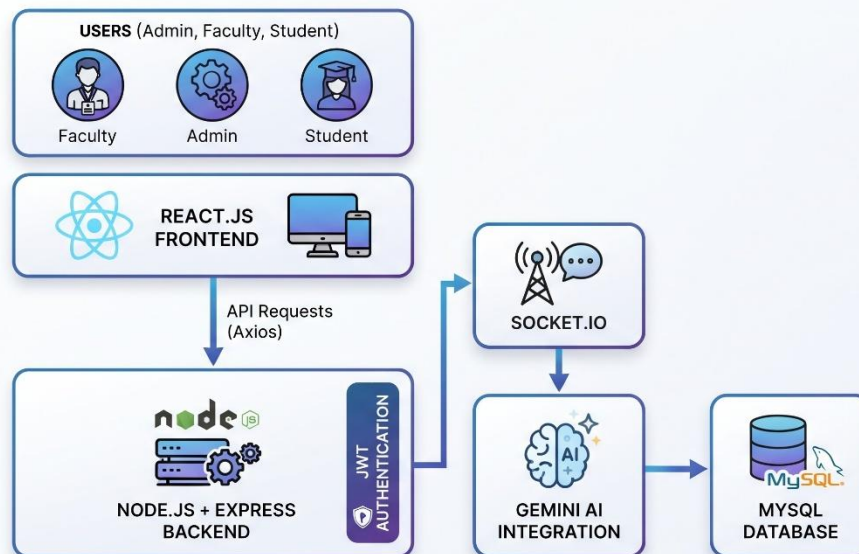
Faculty Workflow - CampusConnect AI



Administrator Workflow - CampusConnect AI



CAMPUSCONNECT AI: SOFTWARE ARCHITECTURE DIAGRAM



AI-Powered Smart Search Workflow - CampusConnect AI

